

Autoencoder-based Visual RL in CarRacing-v3

Bachelor's Degree Programme in Applied Computer Science and Artificial Intelligence

AI Lab: Computer Vision – Signal Processing – Natural Language Processing

Andrea Catino (catino.2154867@studenti.uniroma1.it)

A.Y. 2025/2026



SAPIENZA
UNIVERSITY OF ROME



Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › Proposed Solution and Our Contribution
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Context •

Visual Reinforcement Learning in CarRacing-v3



CarRacing-v3

- Observation
96 × 96 × 3 RGB image
- Action
Continuous control: steering, throttle, brake
- Goal
Learn to drive directly from visual observations
- The agent receives no explicit information about road position, direction or track boundaries.



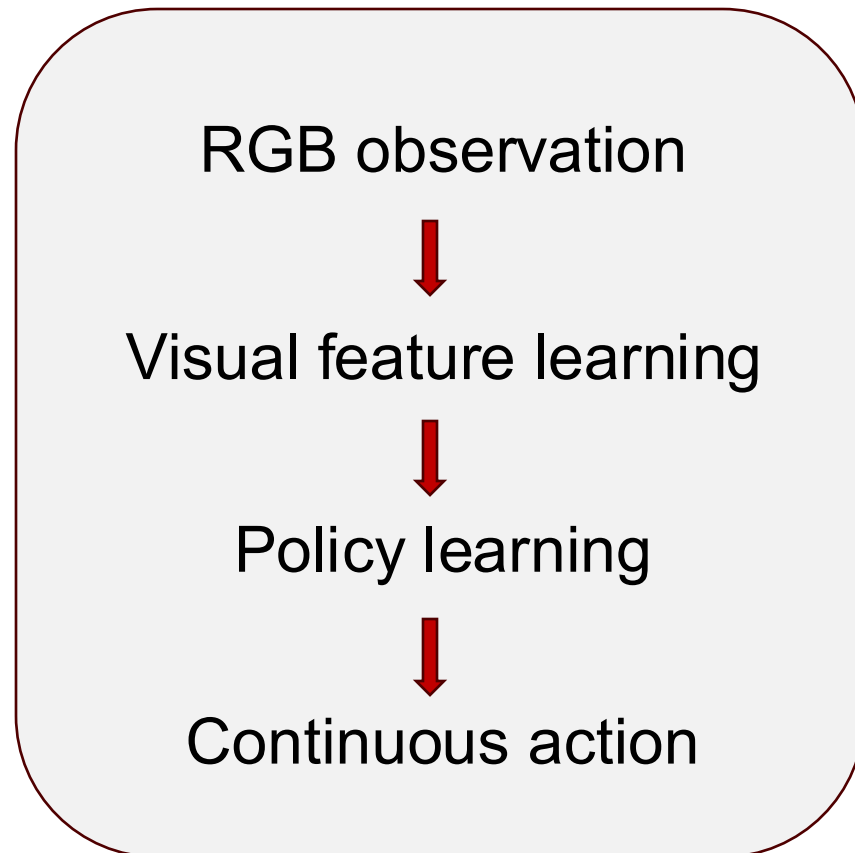
Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › Proposed Solution and Our Contribution
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Learning Perception and Control Simultaneously

Raw-pixel learning



27,648 pixel values → 3 control values

Main challenges

- Visual features are learned only through reward
- Perception and control are optimized simultaneously
- Training can be sample-intensive and unstable
- Can a pretrained visual representation simplify policy learning?



Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › Proposed Solution and Our Contribution
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Related Work and Gap: Learning Control from Pixels

Approach	Main Idea	Limitations
End-to-end Deep RL DQN, PPO	A CNN learns visual features directly from the reward signal	Perception and control are learned simultaneously
Auxiliary Reconstruction During RL SAC+AE	Reconstruction loss helps the encoder learn compact visual features	Representation and policy are still trained jointly
Contrastive / Augmented RL CURL, DrQ	Contrastive learning or image augmentation improves visual representations	More specialized objectives and training strategies

Mnih et al., 2015 · Kostrikov et al., 2021 (DrQ) · Yarats et al., 2021 (SAC+AE) · Laskin et al., 2020 (CURL)

- Limited evaluation of frozen autoencoder representations

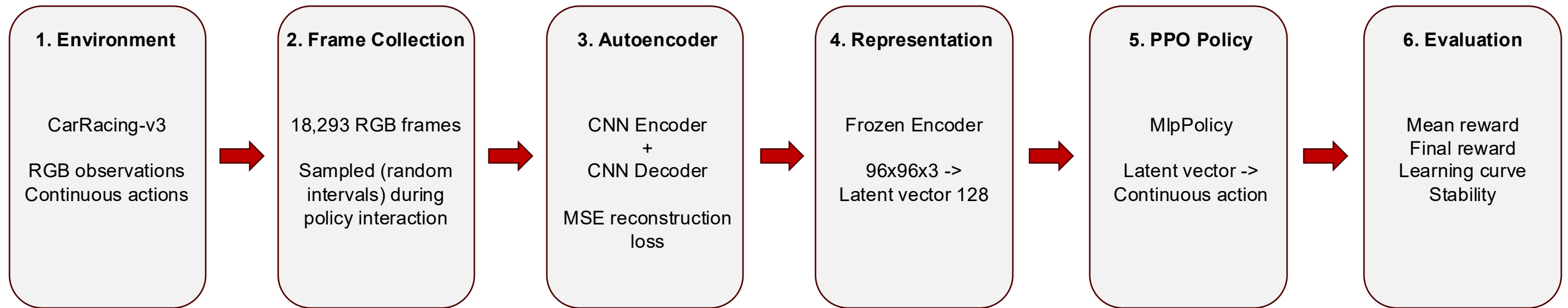


Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › **Proposed Solution and Our Contribution**
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Offline Visual Pretraining



Phase 1: Representation learning

- Reconstruction objective
- No reward signal

Phase 2: Policy learning

- Frozen encoder
- PPO trained only on latent vectors



Offline Visual Pretraining

Contribution

- **Complete Pipeline:** Frame collection, autoencoder training, latent wrapper and PPO integration
- **Controlled Benchmark:** Raw and latent PPO evaluated with the same experimental protocol
- **Reproducibility:** Three training seeds, fixed evaluation seeds and saved configurations
- **Critical Analysis:** Sample efficiency, stability, final performance and failure modes

The contribution is an interpretable and reproducible experimental comparison, not a new RL algorithm.



Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › Proposed Solution and Our Contribution
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Dataset and Autoencoder Architecture

Frame dataset

18,293 PNG frames

96x96 RGB

90% train / 10%
validation

Normalized to [0, 1]

Encoder

3 x 96 x 96

32 x 48 x 48

64 x 24 x 24

128 x 12 x 12

256 x 6 x 6

Latent 128

Decoder

Latent 128

256 x 6 x 6

128 x 12 x 12

64 x 24 x 24

32 x 48 x 48

3 x 96 x 96

Training

MSE loss

Adam

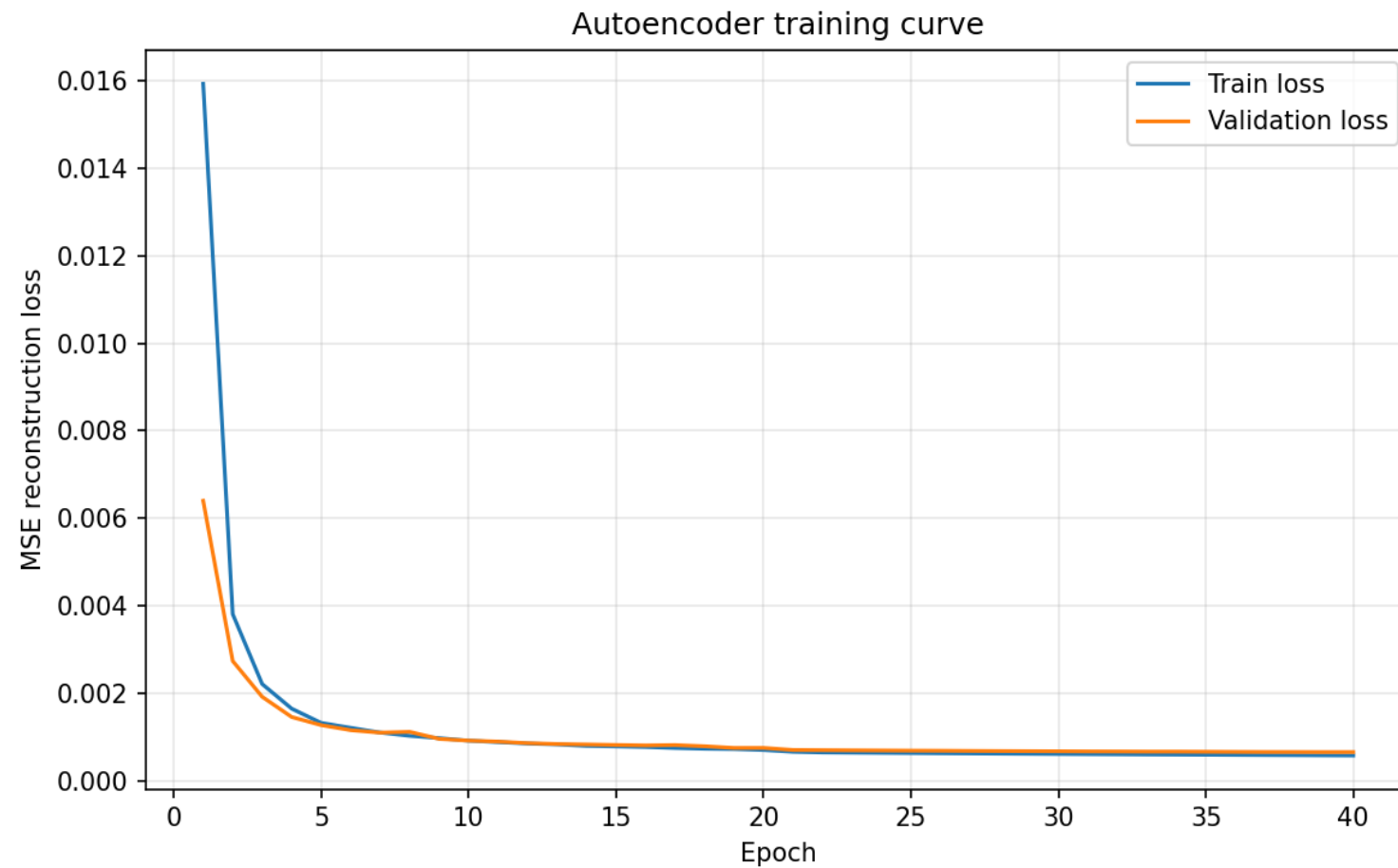
Batch 128

40 epochs

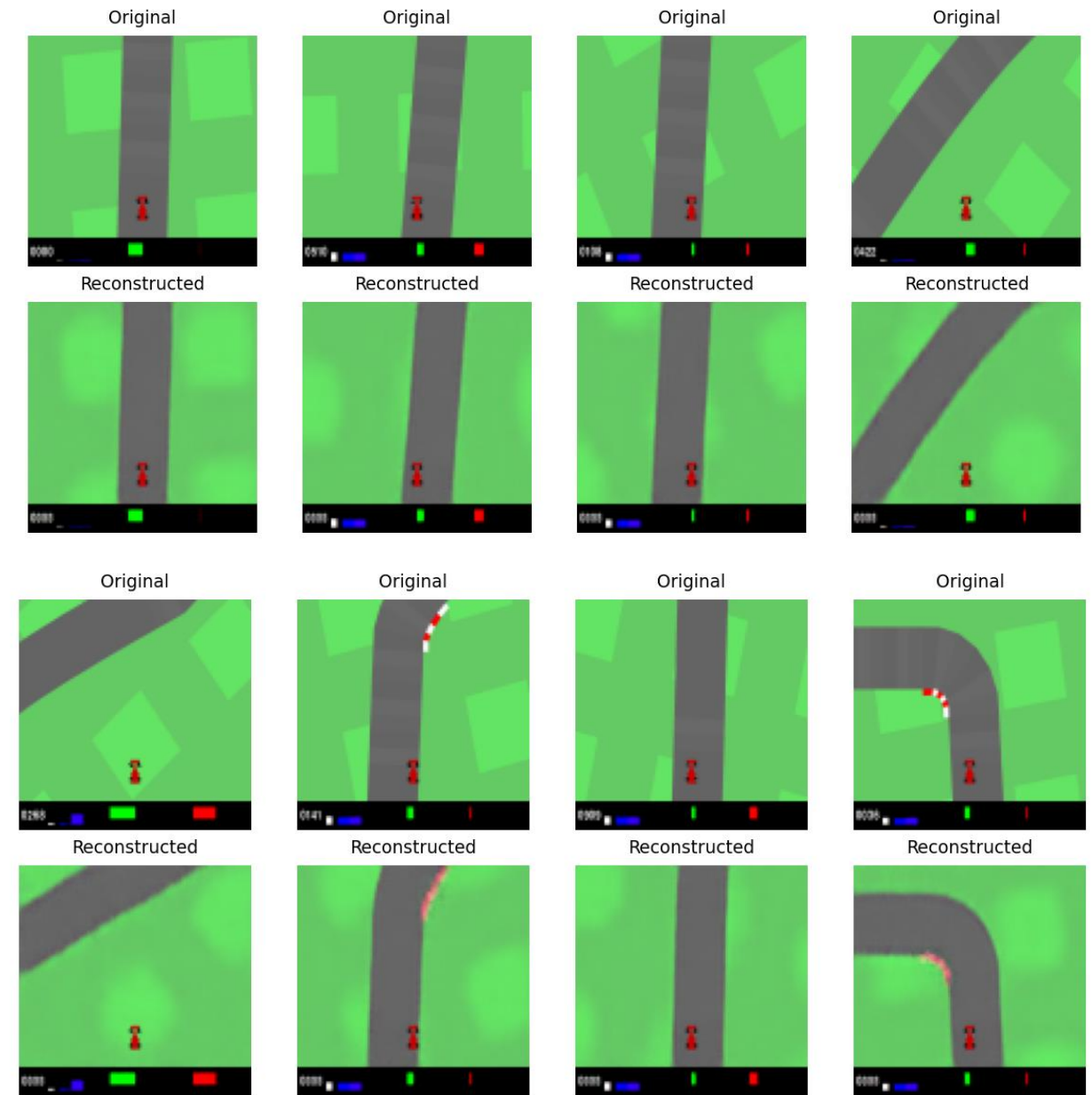
3.75M parameters



Autoencoder Results



Final losses
Train: 0.000581
Validation: 0.000659



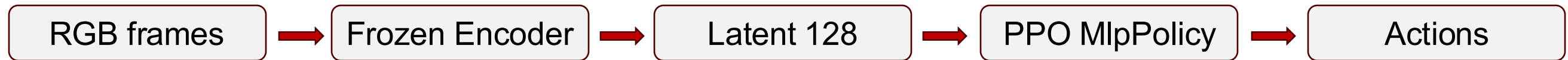


Controlled Experimental Protocol

RAW-PIXEL PPO



LATENT PPO



Shared configuration

- PPO algorithm
- Learning rate: 1e-4
- 2 parallel environments
- 2,000,000 timesteps per run
- Training seeds: 42, 123, 456
- Evaluation seed: 10,000
- 5 deterministic evaluation episodes
- Evaluation every 20,000 environment timesteps



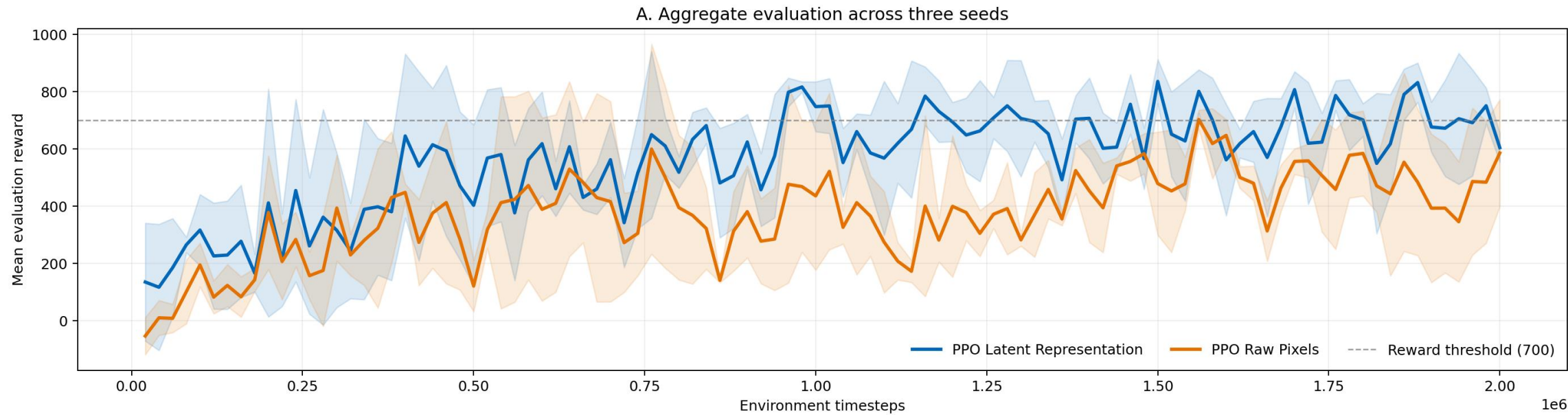
Evaluation Metrics

- **Best Evaluation Reward:** Highest mean reward during training
- **Final-Window Reward:** Mean of the final 5 evaluations
- **Normalized AUC:** Overall performance across the learning curve
- **Steps to Reward 700:** Sample-efficiency threshold
- **Across-Seed Variability:** Mean and standard deviation over 3 training seeds

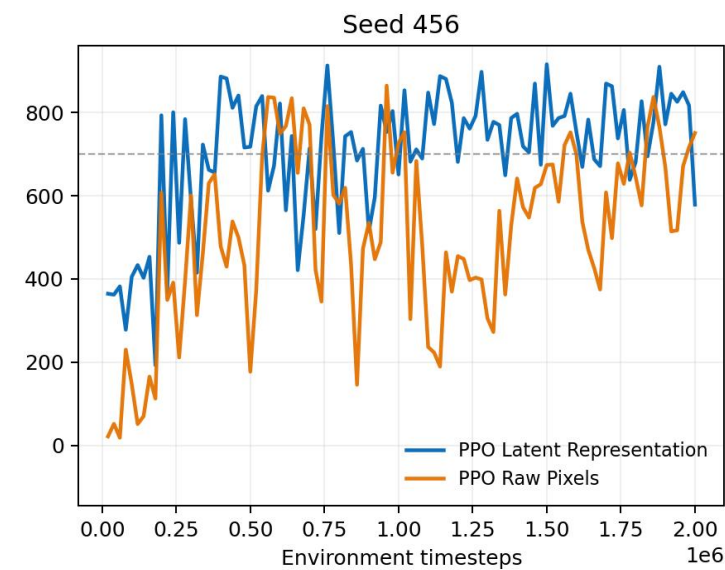
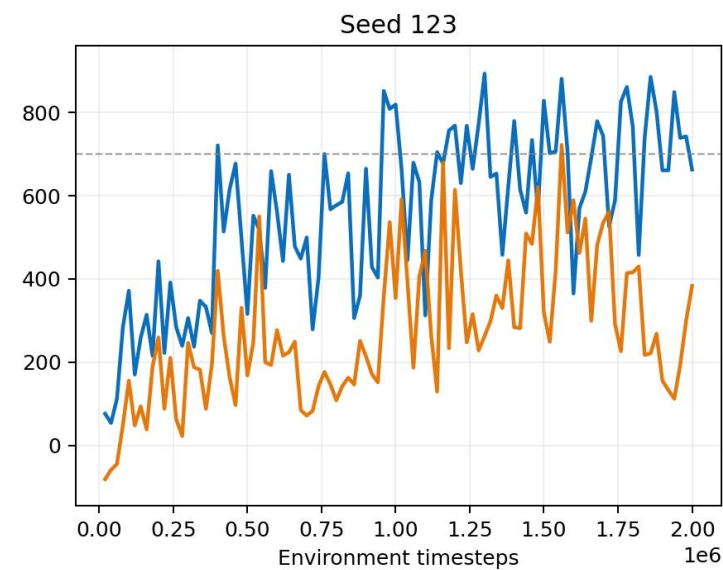
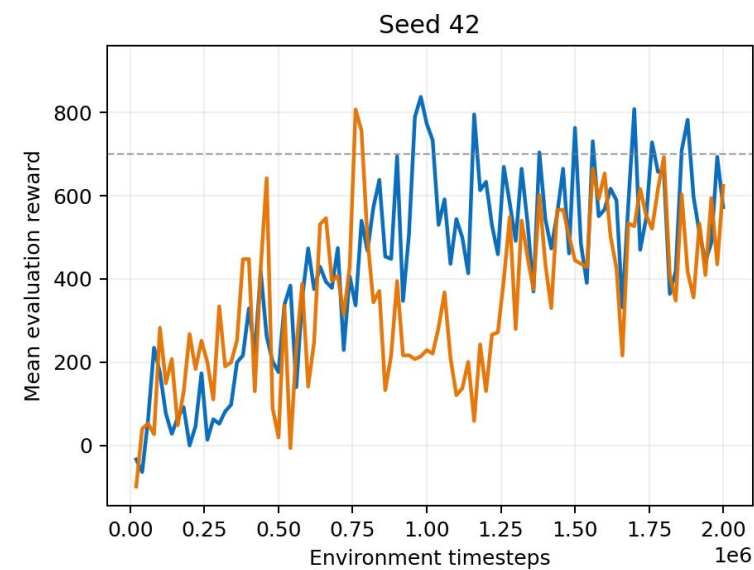


PPO Learning Curves

PPO Evaluation Results Across Three Training Seeds



- Latent PPO achieves higher reward across most of training
- The reward-700 threshold is reached earlier
- Both methods remain variable across seeds





PPO Results Summary

Metric	Raw PPO	Latent PPO	Difference
Best evaluation reward	798.1 ± 71.7	882.4 ± 40.5	+10.6%
Final 5-eval mean	459.2 ± 211.3	685.1 ± 127.4	+49.2%
Normalized AUC	377.8 ± 120.0	567.5 ± 134.5	+50.2%
Steps to reward 700	960k ± 529k	520k ± 394k	-45.8%

- Mean ± standard deviation across 3 training seeds
- Latent PPO outperformed raw-pixel PPO on all predefined aggregate metrics.



Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › Proposed Solution and Our Contribution
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Critical Discussion

What Worked

- Latent PPO outperformed raw PPO on all aggregate metrics
- Reward 700 was reached with 46% fewer timesteps
- Final-window reward improved by 49%
- The advantage appeared across all three seeds for AUC and best reward

What The Results Mean

- The results suggest that reconstruction pretraining captured useful driving information
- Dimensionality was reduced from 27,648 values to 128
- The frozen representation appears to have simplified policy learning
- Separating perception and control was beneficial in this experiment
- The frozen autoencoder improved learning, but it did not eliminate PPO variability.



Limitations

- **Limited Statistical Evidence:** Only three training seeds were evaluated
- **Dataset Bias:** Frames were collected from states visited by previous PPO policies
- **Reconstruction Objective:** MSE optimizes pixel similarity, not control-relevant information
- **Visual Input:** CarRacing observations include the simulator HUD
- **Experimental Scope:** One environment, one latent dimension and one main learning rate
- **Frozen Encoder:** The representation cannot adapt during policy learning



Table of Contents

- › Context
- › Problem and Motivation
- › Related Work and Gap with Respect to the State of the Art
- › Proposed Solution and Our Contribution
- › Dataset and Experimental Protocol
- › Critical Discussion and Limitations
- › Conclusion and Future Work



Conclusion

Does a frozen autoencoder representation improve PPO training from visual observations?
Yes, within the tested CarRacing-v3 setting and experimental budget.

+10.6%
Best evaluation reward

+49.2%
Final-window reward

-45.8%
Timesteps required to reach
reward 700

+50.2%
Normalized learning-curve
AUC

A compact frozen representation improved sample efficiency and overall performance, but PPO remained variable across training seeds.



Future Work

- **Task-Aware Representation Learning:** Fine-tune the encoder during PPO or combine reconstruction and RL losses
- **Better Visual Objectives:** Contrastive learning, augmentation or sharper reconstruction losses
- **Broader Evaluation:** More seeds, different latent dimensions and additional visual-control environments
- **Robustness:** Noise, blur, brightness changes, occlusions and domain randomization
- **Observation Preprocessing:** Remove the HUD and focus the encoder on the road scene



THANK YOU FOR LISTENING!
ANY QUESTIONS?

